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$$\int \frac{dx}{e^{\frac{x}{2}} + e^{\frac{x}{3}}}$$

$$t = e^{\frac{x}{6}} \Rightarrow dx = \frac{6}{t}dt$$

$$\Rightarrow \int \frac{dt}{t^3 + t^2} \cdot \frac{6}{t} = \int \frac{6dt}{t^3(t+1)}$$

$$= \frac{A}{t^3} + \frac{B}{t^2} + \frac{C}{t} + \frac{D}{t+1}$$

$$A = \left.\frac{6}{t+1}\right|_{t=0} = 6$$

$$D = \left.\frac{6}{t^3}\right|_{t=-1} = -6$$

$$\frac{6}{t^3(t+1)} - \frac{6}{t^3} = \frac{-6}{t^2(t+1)}$$

$$\Rightarrow B = \left.\frac{-6}{t+1}\right|_{t=0} = -6$$

$$\frac{-6}{t^2(t+1)} + \frac{6}{t^2} = \frac{6}{t(t+1)}$$

$$\Rightarrow C = \left.\frac{-6}{t+1}\right|_{t=0} = -6$$

$$\Rightarrow 6\int \frac{1}{t^3}dt - 6\int \frac{1}{t^2}dt + 6\int \frac{1}{t}dt - 6\int \frac{1}{t+1}dt$$

$$= \frac{6t^{-2}}{-2} - \frac{6t^{-1}}{-1} + \ln|t| - 6\ln|t+1| + C$$

$$= x - 3e^{\frac{-x}{3}} + 6e^{\frac{-x}{6}} - 6\ln|e^{\frac{-x}{6}} + 1| + C$$

References